

	20	Randomized Block Design (RBD)		
	21	Latin Square Design (LSD).		
	22	Missing plot technique,		
	23	Comparison of Efficiency, Model Adequacy Checking		
	Sections from References:			
V	Open Ended Module: Practical Applications		30	
	1	Designing Experiments, Hands on Using R, Practical Interpretation of Results. Practical problems of ANOVA		
	Sections from References:			
Books and References:				
1. S.C. Gupta & V.K. Kapoor: Fundamentals of Applied Statistics, Sultan Chand & Sons				
2. M.N. Das & N. Giri: Design of Experiments, New Age International				
3. Douglas C. Montgomery: Design and Analysis of Experiments, Wiley and Sons				
4. John Lawson: Design, and Analysis of Experiments with R, Chapman and Hall				

Programme	B. Sc. Statistics				
Course Code	STA6CJ303				
Course Title	Stochastic Processes				
Type of Course	Major				
Semester	VI				
Academic Level	300-399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites					
Course Summary Objective	Objective make students aware of random process, behaviour stationary non stationary discrete continuous indexed process transition probabilities markovian behaviour				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the basic probability concepts including conditional probability, generating functions and Laplace transform	U	C	Instructor-created exams / Quiz
CO2	Student will define and understand the concept of stochastic processes	Ap	P	Practical Assignment / Observation of Practical Skills
CO3	CO3 Student will be able to know and	Ap	P	Seminar

	identify different types of stochastic process based on state space and time space.			Presentation / Group Tutorial Work
CO4	Students will analyse and model systems using discrete-time Markov chains, applying concepts such as transition probabilities, stationary distributions, and limiting behaviour	U	C	Instructor-created exams / Home Assignments
CO5	Develop proficiency in modelling systems with Poisson processes, recognizing their properties and applications across various domains	Ap	P	One Minute Reflection Writing assignments
CO6		Ap	P	Viva Voce
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48 +12)	Marks (70)
I			10	15
	1	Introduction to Stochastic Processes(SP)		
	2	Definition of state space and time space		
	3	Classification of SP according to state space and time space,.		
	4	Process with independent increment		
	5	Process with stationary increment		
	Sections from References:			
II			10	20
	6	Markov property		
	7	Markov Chain		
	8	Discrete time Markov Chain(MC),		
	9	Transition probability matrix,		
	10	MC as graph.		
	11	Higher transition probabilities,		
	12	Chapman- Kolmogorov Equation.		
	13	One dimensional random walk (concept only)		
	Sections from References:			
III			20	20
	14	First passage probabilities		
	15	PGF,		
	16	Different types of states, classification of states (Recurrent, transient, ergodic)		
	17	Periodicity, mean ergodic theorem (statement only)		
	18	Class property, stationary distribution, limiting distributions,		

	19	Gambler's ruin problem (concept and construction of tpm only).		
	Sections from References:			
IV			8	15
	20	Continuous time MC,		
	21	Chapman-Kolmogorov equation (statement only),		
	22	Poisson Process		
	23	Inter-arrival time.		
	22	Relationship connecting Poisson Process and distributions (exponential, binomial, uniform and geometric)		
	Sections from References:			
V	Open Ended Module:		12	
	1	Practical problems relating to previous modules.		
	Sections from References:			
Books and References:				
1. Medhi J. (2014) Stochastic Processes. Third Edition, New Age International				
2. Basu A.K. (2003) Introduction to Stochastic Processes, Narosa, New-Delhi.				
3. Cinlar E. (2013) Introduction to Stochastic Processes, Dover Publications, NewYork				
4. Feller W. (1968) Introduction to Probability Theory and its Applications, Vols. I & II, John Wiley, New York.				
5. Karlin S. and Taylor H.M. (1975) A First Course in Stochastic Processes, Second edition, Academic Press, New-York.				
6. Ross S.M. (2014) Introduction to Probability models, Eleventh edition, Academic Press				

SEMESTER VII

Programme	B. Sc. STATISTICS
-----------	-------------------